BARRETT LEMONS

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EDUCATION

Bachelor of Science, Mechanical Engineering Expected May 2030

University of Tennessee — Knoxville, TN

- Chancellor's Honors Program; Cook Grand Challenge Engineering Honors Program
- Recipient: Volunteer, Flagship, and Ned McWherter Scholarships

High School Diploma May 2026

Central High School — Knoxville, TN

- Valedictorian — GPA: 4.60 — ACT: 34
- Tennessee State Seal of Biliteracy (English/Spanish)

CORE COMPETENCIES

- AutoCAD & Civil 3D Design; PLS-CADD (pole / line FEA); 3D Modeling & Visualization
- Field Inspection & Technical Documentation; Safety Compliance & Standards
- Microsoft Office Suite; Python (engineering analysis); Bilingual English / Spanish

PROFESSIONAL EXPERIENCE

Electric Systems Engineering Intern Jun 2025 – Jul 2025; Dec 2025 – Jan 2026

Knoxville Utilities Board (KUB) — Knoxville, TN

- Converted a legacy 2D assembly drawing into a 3D model, enabling quick angle adjustments and eliminating repeated manual redraws for a colleague.
- Designed the relocation of a utility pole and redesign of an adjacent service line for a City of Knoxville greenway project near Fulton High School, using PLS-CADD to run finite element analysis on the affected poles; work was reviewed and approved by a KUB engineer.
- Independently developed two alternative electric service designs, underground and pole-mounted, to power a sign at Central Magnet Academy, iterating through multiple rounds of revision before final review and approval.
- Conducted field inspections, roughly a third of total time, to verify customer complaints and confirm structure locations that GIS records could not reliably provide.
- Created approximately 10 technical drawings and revisions to support cost estimates for utility projects in the $20,000+ range.

Office Assistant Aug 2026 – present

Department of Enrollment Management Communications, University of Tennessee — Knoxville, TN

- Prepare and route outgoing mail and informational packages for prospective-student communications, balancing a 10-hour weekly commitment alongside a full engineering course load.

Server May 2026 – Jun 2026

Holston Hills Country Club — Knoxville, TN

- Built an Excel spreadsheet to track ABC license compliance, helping prevent audit issues.
- Served guests in a fast-paced country club dining room, balancing multiple tables while maintaining service standards.

Lifeguard Apr 2024 – Aug 2025

Holston Hills Country Club; City of Knoxville — Ed Cothren Pool

- Monitored pool areas, enforced facility rules, and conducted routine equipment inspections for safety compliance.
- Performed 2 water rescues and delivered first aid during emergency safety incidents (Ed Cothren Pool, 2024).

ACTIVITIES

Student Space Technology Association (SSTA) — Hypersonics Team Fall 2026 – present

EF 157 Glider Design Project Aug 2026 – present

- Helping coordinate a 5-person team designing a self-propelled, self-launching glider for a competitive accuracy and payload-delivery tournament; working through trade-offs in wing loading, descent profile, and launch mechanism design.

UTK Ballroom Dance Team — Member